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DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956

SCHOOL OF ENGINEERING COIMBATORE

Department of Electrical and Electronics Engineering

19CSE362 · Fundamentals of Soft Computing · Project Review

Explainable ANN–ANFIS Hybrid for Renewable-Aware Task Offloading in Solar-Powered IoT Nodes Using a Digital Twin

Four soft-computing algorithms in one model

ALGORITHM 1

ANN

Multilayer perceptron

- Learns how the WiFi link and the server behave
- Predictor 6 → 24 → 16 → 2: offload time and energy
- Classifier 9 → 24 → 16 → 1: the "ANN only" baseline

R^2 0.989 · alone: 96.7 %

ALGORITHM 2

Fuzzy logic

Mamdani inference

- Reasons with words such as Low, Fair and Heavy
- 4 inputs with 3 words each
- 12 hand-written IF–THEN rules, centroid output

Expert baseline · 92.5 %

ALGORITHM 3

ANFIS

Adaptive neuro-fuzzy system

- Fuzzy rules that learn from data
- 5 layers with bell-shaped curves, 36 rules
- Least squares + gradient descent

Alone: 93.7 %

ALGORITHM 4 · PROPOSED

ANN + ANFIS hybrid

Predict, then reason

- The ANN predicts what offloading will cost
- ANFIS turns that prediction into a decision
- Every decision shows the rules that fired

96.8 % · lowest energy

How it works: from digital twin to decision

Digital twin: a physics simulator, one step = one minute

Solar PV
1 W panel; sunlight and clouds change through the day

→

Battery
12 Wh; charged by solar, drained by the node

→

ESP32 node
240 MHz; one IoT task arrives every minute

→

WiFi / MQTT
signal, speed and retries change with distance

→

Edge server
3 GHz; much slower when it is busy

1

What the ESP32 knows before deciding

battery level · task size and type · its own local time and energy · last measured signal · last server load

2

Dataset: 10,080 tasks × 38 features

each task labelled with the cheaper of the two exact costs (the oracle): 54 % local, 46 % offload

trains the ANN and ANFIS (teacher)

3

Algorithm 1 ANN predictor

6 → 24 → 16 → 2 neurons; predicts the offload time and energy

4

Algorithm 3 ANFIS · 36 rules

4 inputs, 5 layers; learned fuzzy rules give an offload score

Algorithm 4 · ANN + ANFIS hybrid (proposed)

5

Decision

score above 0.5 → OFFLOAD to the edge, otherwise run LOCAL

closed loop: the chosen option's energy is drawn from the battery

6

Evaluation on 11 unseen weeks

against Fuzzy (Algorithm 2), ANN alone (Algorithm 1) and ANFIS alone (Algorithm 3), with a Wilcoxon test

7

Explainable dashboard

shows every decision and the rules that fired

96.8 %
agreement with the best possible decision (hybrid, 11 unseen weeks)

11 / 11
weeks with less energy than each algorithm on its own

36
learned fuzzy rules that explain every decision

20 / 20
engineering checks pass on the digital twin